

8. Database Schema Design

8.1 Design Principles

- Third Normal Form (3NF) with selective denormalization for read-heavy analytics queries
- UUID v4 primary keys on all tables for distributed-safe generation
- Audit columns on every table: created_at, updated_at, created_by, updated_by
- Soft deletes: is_deleted (boolean) + deleted_at (timestamp); no physical deletes
- All schema changes via Alembic migration scripts; no manual DDL in production
- PII columns (aadhaar, pan, bank_account) encrypted via Azure SQL column-level encryption
- ENUM types used for status fields to enforce data integrity at the database level

8.2 Core Entity Tables

8.2.1 employees

Central entity for all employee data. Referenced by every other module.

Column	Type	Null	Description
id	UUID (PK)	No	Auto-generated unique identifier
employee_code	VARCHAR(20)	No	Auto-generated sequential code (e.g., EMP001)
entra_object_id	VARCHAR(128)	No	Azure Entra ID object ID for SSO linkage
first_name	VARCHAR(100)	No	Employee first name
last_name	VARCHAR(100)	No	Employee last name
email	VARCHAR(150)	No	Official email address (unique)
phone	VARCHAR(15)	Yes	Contact number
date_of_birth	DATE	No	Date of birth
date_of_joining	DATE	No	Date of joining the organization
date_of_exit	DATE	Yes	Date of exit (null if active)
department_id	UUID (FK)	No	Reference to departments table
designation_id	UUID (FK)	No	Reference to designations table
reporting_manager_id	UUID (FK)	Yes	Self-referencing FK to employees
employment_status	ENUM	No	active, on_notice, exited, suspended
gender	ENUM	No	male, female, other
aadhaar_encrypted	VARBINARY	Yes	Encrypted Aadhaar number
pan_encrypted	VARBINARY	Yes	Encrypted PAN number
bank_account_encrypted	VARBINARY	Yes	Encrypted bank account number
bank_ifsc	VARCHAR(11)	Yes	Bank IFSC code
profile_photo_url	VARCHAR(500)	Yes	SharePoint Online URL
is_deleted	BIT	No	Soft delete flag (default: 0)
created_at	DATETIME	No	Record creation timestamp

updated_at	DATETIME	No	Last update timestamp
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8.2.2 departments

Column	Type	Null	Description
id	UUID (PK)	No	Department identifier
name	VARCHAR(100)	No	Department name (e.g., Engineering, HR, Finance)
head_id	UUID (FK)	Yes	Reference to employees (department head)
parent_department_id	UUID (FK)	Yes	Self-referencing FK for hierarchy

8.2.3 designations

Column	Type	Null	Description
id	UUID (PK)	No	Designation identifier
title	VARCHAR(100)	No	Designation title (e.g., Senior Engineer, HR Manager)
level	INT	No	Hierarchy level (1=entry, 10=CXO)